

RAHUL MITRA

CIVIC TECH SOLUTION ARCHITECT PROFILE

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PROFILE SUMMARY

Solution architect and full-stack engineer building technology for social good. Delivers production civic and healthcare systems end-to-end - cloud architecture, Singpass-integrated platforms, GIS, and applied AI - turning community and stakeholder needs into deployed software that reaches vulnerable groups.

EDUCATION

National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Software Engineering; Artificial Intelligence; Simulation; Statistics for Engineering Applications; Operations Research; 3D Computer Vision
- Certificate of Outstanding Performance in 3D Computer Vision (CS4277) - top student in a class of 24, NUS School of Computing.

Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

WORK AND INTERNSHIP EXPERIENCE

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd

Jun 2026 - Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.
- Strengthen monitoring and process structure across healthcare supply chains with AI, helping ensure reliable product availability for vulnerable patient groups.

Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd

Jan 2026 - Jun 2026

- Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules.
- Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.

Software Engineer (Citizen Developer), People's Association

Sep 2025 - Present

- Built an end-to-end digital garden-plot balloting platform (admin and public-facing frontends, backend, GIS mapping, and deployment), now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks.
- Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders; awarded the S\$20,000 Sparks Community Innovation Fund.

SELECTED PROJECTS AND COMPETITIONS

Azure APC Web Simulator

2026

- Designed a cloud-hosted process-control simulation surface with Azure deployment, Docker packaging, CI/CD pipelines, and manufacturing stakeholder feedback loops.

Volt Pulse SG - SMU Hack For Cities Finalist

Jan 2026

- Reached Top 8 of 50+ teams with an agentic AI platform for HDB energy-cost tracking using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

AgeWellLah.AI - NUS HealthHack 2025

2025

- Engineered an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, Flask, OAuth2-secured patient profiles, and PDPA-aware design choices.

EthosLens - LifeHack 2025

2025

- Developed backend workflows for an AI research engine that scores brand ethics and sustainability practices using FastAPI, Python, LangChain, and automated evidence retrieval.

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Solution Architecture; Cloud (Azure, Docker, CI/CD); Full-Stack (React, TypeScript, FastAPI, Flask); Singpass Login & Myinfo (OIDC); RAG & AI Agents; GIS Mapping; API Design; Stakeholder Discovery; Accessibility; Rapid Prototyping.

- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Docker for the Absolute Beginner - container fundamentals and deployment workflows.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Sustained a year of weekly senior engagement, deepening empathy for accessibility, healthcare, and human-centered community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.